

Maharashtra State Board Of Technical Education, Mumbai																										
Learning and Assessment Scheme Of Diploma Courses for Working Professionals																										
Programme Name					: Diploma In Civil Engineering (Working Professional)																					
Programme Code					: PCE					With Effect From Academic Year					: 2024-25											
Duration of Programme					: 6 Semester																					
Semester					: Fourth					Scheme					: F											
Sr No	Course Title	Abbreviation	Course Type	Course Code	Diploma Awarding Course	Total IKS Hrs for Sem.	Learning Scheme					Credits	Assessment Scheme													Total Marks
							Actual Contact Hrs./Week			Self Learning (Activity/ Assignment /Micro Project)	Notional Learning Hrs /Week		Paper Duration (hrs.)	Theory			Based on LL & TL				Based on Self Learning					
							CL	TL	LL								Practical									
														FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA				
																Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
(All Compulsory)																										
1	RAILWAY, BRIDGE AND TUNNEL ENGINEERING	RBT	DSC	314312	No	3	4	-	-	2	6	3	3	30	70	100	40	-	-	-	-	25	10	125		
2	STRENGTH OF MATERIALS	SOM	DSC	313308	No	1	4		2		6	3	3	30	70	100	40	25	10	-	-	-	-	125		
3	GEOTECHNICAL ENGINEERING	GTE	DSC	314315	No	1	3	-	2	3	8	4	3	30	70	100	40	25	10	25#	10	25	10	175		
4	BUILDING PLANNING & DRAWING WITH CAD	BDC	SEC	313009	No	1	2		4	2	8	4	-	-	-	-	50	20	50#	20	25	10	125			
Total						6	13	0	8	7		14		90	210	300		100		75		75		550		
Abbreviations : CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA -Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment																										
Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination																										
Note :																										
1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.																										
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.																										
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.																										
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks																										
5. 1 credit is equivalent to 30 Notional hrs.																										
6. * Self learning hours shall not be reflected in the Time Table.																										
7. * Self learning includes micro project / assignment / other activities.																										
Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project./Community (INP) , AbilityEnhancement Course (AEC) , Skill Enhancement Course (SEC) , GenericElective (GE)																										

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312**

Programme Name/s	: Civil Engineering/ Civil & Rural Engineering/ Construction Technology/ Civil & Environmental Engineering/
Programme Code	: CE/ CR/ CS/ LE
Semester	: Fourth
Course Title	: RAILWAY, BRIDGE AND TUNNEL ENGINEERING
Course Code	: 314312

I. RATIONALE

Railway, Bridge and Tunnel Engineering is an important aspect of Civil Engineering as they are very crucial in shortening the distance of travel. Efficient and Effective network of different modes of transportation plays an important role in the Nation's economic progress and its integration. The basic requirements of efficient transportation are speed, safety and comfort. This course is intended to develop the basic skills related to investigation, surveys, alignment, construction and maintenance of Railway, Bridge, and Tunnels.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Execute the construction and maintenance of railways, bridges and tunnels.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Identify the relevant components of Railway Tracks.
- CO2 - Maintain the given Railway Track.
- CO3 - Maintain the given type of bridge through due inspection.
- CO4 - Suggest the relevant method of constructing a tunnel in the given strata.
- CO5 - Supervise the construction of tunnels including maintenance activities.

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312****IV. TEACHING-LEARNING & ASSESSMENT SCHEME**

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL	FA-TH	SA-TH			Total		FA-PR		SA-PR		SLA				
													Max	Min	Max	Min	Max	Min	Max	Min	
314312	RAILWAY, BRIDGE AND TUNNEL ENGINEERING	RBT	DSC	4	-	-	2	6	3	3	30	70	100	40	-	-	-	-	25	10	125

Total IKS Hrs for Sem. : 3 Hrs

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312****V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe the development of Indian railways till date. TLO 1.2 Show the components of railway track in the given cross section of track. TLO 1.3 Suggest the types of sleepers provided for the specified railway track with justification. TLO 1.4 Propose the relevant type of ballast to be provided in specified railway track with justification. TLO 1.5 Identify the fixtures with fastening provided in the given rail section.	Unit - I Introduction to Railway Engineering 1.1 History of development of railways in India (IKS) ,Railway: Zones of Indian railways, Merits and demerits of roadway and railway, Introduction to Metro and Mono rail, Bullet Train. 1.2 Components of railway track: Rails , ideal requirements of railway track , types of Rails ,Rail Gauge- types, factors affecting selection of a gauge. tilting of rails and coning of wheels. Rail Joints : Necessity, types, requirements of welded joints. Creep of rail: Definition, causes and prevention of creep. 1.3 Sleepers : Requirement, functions and types, sleeper density 1.4 Ballast : requirement, function, types, suitability. 1.5 Rail fixtures and fastenings: fish plate, spikes, bolts, keys, bearing plates, chairs , types of anchors and anti-creepers.	Model Demonstration Video Demonstrations Lecture Using Chalk-Board Site/Industry Visit Case Study

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Fix the alignment of given railway track laid on typical terrain. TLO 2.2 Draw the cross section of the track showing its geometric elements with neat labels. TLO 2.3 Explain the track geometric components with its importance in its design. TLO 2.4 Explain with sketches the concept of turn outs, points, and crossings w.r.t railway track. TLO 2.5 Propose the relevant type of station with its salient parameters considered in its site selection. TLO 2.6 Justify the necessity of station yard in railway engineering. TLO 2.7 Maintain the track in the capacity of the permanent way inspector.	Unit - II Track Geometrics 2.1 Alignment: Factors governing rail alignment. 2.2 Cross sections of Track : Important technical terms- permanent land width/right of way, formation width, side slopes, side drains. Standard cross section of single and double line in cutting and in embankment. 2.3 Railway Track Geometrics: types and factors affecting Gradient , curves , grade compensation , super elevation- limits of Super elevation on curves , cant deficiency (No numerical in question-paper). 2.4 Branching of Tracks: Points and crossings: Turn out- left and right-hand turnout, components, and their functions ,important technical terms ,track junctions- crossovers, scissor cross over, diamond crossing, track triangle. 2.5 Railway Station : Purpose , requirement of railway station , factors affecting site selection for railway station, important technical terms , types of railway station. 2.6 Station yard: Function , Classification- Passenger, goods, locomotive and marshalling yards, drawbacks of marshalling yards. 2.7 Track Maintenance: Necessity , Classification , Tools required for track maintenance with their function , Organization of track maintenance , Duties of permanent way inspector, gang mate and key man.	Model Demonstration Video Demonstrations Case Study Presentations Lecture Using Chalk-Board Site/Industry Visit

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Elaborate the typical features of major important bridges in India. TLO 3.2 Suggest the relevant type of bridge based on available data. TLO 3.3 Explain Factors affecting Site selection of given type of bridge. TLO 3.4 Explain with sketch Important technical terms related to a bridge. TLO 3.5 Explain with neat sketches the given component of bridge. TLO 3.6 Suggest the relevant type of bridge to be used in the given situation. TLO 3.7 Undertake the inspection of bridge during Pre and post monsoon period. TLO 3.8 Maintain the given type of bridge.	Unit - III Bridge Engineering 3.1 History of development of bridges in India (IKS) 3.2 Classification of bridges: according to span, purpose, material, life, alignment, H.F.L, Loading, level of bridge floor. 3.3 Site selection and investigation Factors affecting and controlling: Site For Bridge, Bridge Alignment. 3.4 Important technical terms: Waterway, Economic Span , Afflux ,Scouring , Erosion, Freeboard , Cut Water ,Ease Water, Apron 3.5 Component parts of bridge: Function, requirement, and types- Pier , Abutment , Wing Wall , Foundation ,Bearing 3.6 Types of Bridges: Causeway: Flush, low level, and high-level causeway. RCC Bridges , Pre-stressed bridge: Advantage & dis-advantages, Culvert: Types- Arch, Open or slab, Pipe and box 3.7 Inspection of bridges: General points to be observed, Pre and post monsoon inspection. 3.8 Maintenance of bridges: types - routine and special Maintenance.	Model Demonstration Video Demonstrations Case Study Presentations Lecture Using Chalk-Board Site/Industry Visit

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Summarize the typical features of major important tunnels in India. TLO 4.2 Identify the type of the tunnel from the given sketch. TLO 4.3 Explain the criteria for selection of the tunnel for given situation with justification. TLO 4.4 Describe the process of shifting the alignment inside the tunnel through shaft. TLO 4.5 Suggest the relevant method of constructing the tunnel in the given terrain.	Unit - IV Tunnel Engineering 4.1 History of development of tunnels in India (IKS). 4.2 Classification of tunnels: according to purpose, conveyance, strata through which tunnel passing, alignment, shape, and size of tunnels. 4.3 Tunnels: Tunnel investigations and surveying, Cross sections for highways and railways. 4.4 Tunnel Shaft : its purpose and construction. 4.5 Methods of tunnelling in soft rock: Needle Beam method, Fore-Poling method, Line Plate method, Shield method.	Model Demonstration Video Demonstrations Case Study Presentations Lecture Using Chalk-Board Site/Industry Visit

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Suggest the relevant method of constructing the tunnel in the available ground strata. TLO 5.2 Select the relevant type of drilling machine for the given strata. TLO 5.3 Describe the process of lining in the given tunnel in the given situation with justification. TLO 5.4 Justify the need to provide the provision for ventilation and drainage in the tunnel. TLO 5.5 Describe the procedure of maintenance of the given tunnel.	Unit - V Construction and Maintenance of Tunnels 5.1 Methods of Tunnelling in Hard Rock: Full-face method, Heading and bench method, drift method, New Austrian Tunnelling Method (NATM). 5.2 Drilling Equipment: TBM Tunnel Boring Machine, drills and drills carrying equipment's, Types of explosives used in tunnelling. 5.3 Tunnel Lining: Purpose, factors affecting type of lining, and methods. 5.4 Tunnel Ventilation and Drainage: Purpose and methods. 5.5 Tunnel Maintenance: Purpose and Methods.	Model Demonstration Video Demonstrations Case Study Presentations Lecture Using Chalk-Board Site/Industry Visit

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES : NOT APPLICABLE.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

MSBTE Approval Dt. 21/11/2024

Semester - 4, K Scheme

RAILWAY, BRIDGE AND TUNNEL ENGINEERING

Course Code : 314312

Micro project

- Prepare report on Railway Zones in India .
- Prepare model of a bridge/Tunnel to demonstrate the relevant concepts.
- Prepare models of different gauges used in railways.
- Collect the details of new technologies of tunnel excavation and prepare the report.
- Collect the information relevant to transportation engineering about ongoing and completed Railway/Bridge/Tunnel projects. (Minimum 3)
- Role of Indian Railway (IR), MSRDC, NHAI and IRC in development and construction of Railways, Tunnels and Bridges.
- Prepare a report on Bullet Train, Mono rail, Metro Rail project.
- Summarize the salient features of relevant IS codes used in this course in the form of a report.

Assignment

- Inspect nearby Railway Track /Bridge/Tunnel (any one) to enumerate the defects if any and prepare the report suggesting remedial measures for ensuring its stability.
 - Draw the standard cross section of single line and double line railway on embankment and in cutting.
 - List the advanced equipment's/machineries and materials required for preparation of subgrade of railway.
 - Compile the relevant information on project Atal tunnel/Patalpani Rail tunnel with your own comments.
 - Compile the relevant information on project Bandra Worli sea link bridge/Pamban Bridge with your own comments.
 - Visit a nearby Bridge site/Tunnel and prepare a detailed photographic report.
 - Compile the relevant information on project Mumbai to Ahmadabad Bullet Train with your own comments.
 - Prepare a site visit report to the nearby railway station mentioning the details of the type of station, requirements fulfilling the station and any other important findings with required figures and facts.
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RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312****Note :**

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Computer with internet facility	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Introduction to Railway Engineering	CO1	10	4	4	4	12
2	II	Track Geometrics	CO2	18	4	12	6	22
3	III	Bridge Engineering	CO3	14	2	6	6	14
4	IV	Tunnel Engineering	CO4	10	4	4	4	12

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312**

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
5	V	Construction and Maintenance of Tunnels	CO5	8	0	4	6	10
Grand Total				60	14	30	26	70

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Under SLA : Assignment, Microproject (60% Weightage to process and 40% weightage to product), Question and Answer

Summative Assessment (Assessment of Learning)

- Pen and Paper Test (Written Test)

XI. SUGGESTED COS - POS MATRIX FORM

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312**

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	----	----	2	2	----	3			
CO2	3	1	1	2	2	1	3			
CO3	3	2	2	3	2	2	3			
CO4	3	3	3	2	3	2	3			
CO5	3	3	3	3	3	2	3			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	S. C. Saxena , S. P. Arora	A Text Book of Railway Engineering	Dhanpat Rai Publications (p) Ltd.-New Delhi ISBN-13:978-8189928834
2	Bindra S. P.	Elements of Bridge ,Tunnel & Railway Engineering	Dhanpat Rai Publications (p) Ltd.-New Delhi ISBN: 9789383182220, 9383182229

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312**

Sr.No	Author	Title	Publisher with ISBN Number
3	Ahuja & Birdi	Roads, Railways, Bridges and Tunnels Engineering	Standard Book House ISBN: 978-81-89401-33-7
4	Raji A K, K K Babu	Transportation Engineering (Theory and Practice)	AICTE New Delhi ISBN 978-81-960576-1-9
5	N L Arora	Transportation Engineering	New India Publishing House, New Delhi
6	R. Srinivasan	Harbour, Dock and Tunnel Engineering	Charotar Publishing House Pvt. Ltd. ISBN-13 978-9385039195

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://indianrailways.gov.in/	Indian Railway Zones (IKS)
2	https://iricen.gov.in/iricen/BooksList.jsp	IRICEN Books on Railway and Bridge Engineering
3	https://nhsrcl.in/en/home	National High Speed Rail Corporation Limited (Bullet Train)
4	https://msrdc.in/Site/Common/ProjectListDetails.aspx?ID=56&MainId=18	Versova-Bandra Sea Link Project by MSRDC
5	https://marvels.bro.gov.in/AtalTunnel	Atal Tunnel, Rohtang
6	https://archive.nptel.ac.in/courses/105/105/105105216/	Bridge Engineering video lectures by NPTEL
7	https://nptel.ac.in/courses/105107123	Railway Engineering video lectures by NPTEL
8	https://mmrda.maharashtra.gov.in/projects/transport/metro-line-1/overview	Mumbai Metropolitan Region Development Authority

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

RAILWAY, BRIDGE AND TUNNEL ENGINEERING**Course Code : 314312**

Sr.No	Link / Portal	Description
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

STRENGTH OF MATERIALS

Course Code : 313308

Programme Name/s	: Automobile Engineering./ Agricultural Engineering/ Civil Engineering/ Civil & Rural Engineering/ Construction Technology/ Civil & Environmental Engineering/ Mechanical Engineering/ Production Engineering/
Programme Code	: AE/ AL/ CE/ CR/ CS/ LE/ ME/ PG
Semester	: Third
Course Title	: STRENGTH OF MATERIALS
Course Code	: 313308

I. RATIONALE

All civil & mechanical engineering components are subjected to different types of loads and behave in a specific way. Students can able to understand & analyze various types of loads, stresses & strains with regards to the structural behavior of components and materials. This course is a prerequisite for understanding elastic behavior of different engineering materials, structural analysis, machine design, principles and the strengths of various structural elements used in civil & mechanical industries.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Analyze the stresses & strains in the given structural elements using relevant methods.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Calculate the M.I. of the given object using relevant formulae & methods.
- CO2 - Analyze the structural behavior of the given structural components under various loading conditions.
- CO3 - Draw SFD and BMD for the given structural element under given loading conditions.

STRENGTH OF MATERIALS**Course Code : 313308**

- CO4 - Determine the bending and shear stresses in beams under different loading conditions
- CO5 - Analyze the direct & bending stresses in the structural members under eccentric loading conditions.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Paper Duration	Assessment Scheme													
				Actual Contact Hrs./Week			SLH	NLH			Theory	Based on LL & TL				Based on SL		Total Marks						
												Practical												
				CL	TL	LL						FA-TH		SA-TH		Total			FA-PR		SA-PR		SLA	
												Max	Min	Max	Min	Max	Min		Max	Min	Max	Min	Max	Min
313308	STRENGTH OF MATERIALS	SOM	DSC	4	-	2	-	6	3	3	30	70	100	40	25	10	-	-	-	-	125			

STRENGTH OF MATERIALS**Course Code : 313308****Total IKS Hrs for Sem. : 1 Hrs**

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
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5. 1 credit is equivalent to 30 Notional hrs.
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V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
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STRENGTH OF MATERIALS**Course Code : 313308**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Calculate Centroid, Moment of Inertia of Plane lamina and radius of gyration of a given lamina. TLO 1.2 Explain Parallel and Perpendicular axes theorems. TLO 1.3 Calculate Moment of inertia of standard solid shapes and hollow shapes. TLO 1.4 Calculate Moment of inertia of composite plane figures such as I, C, T & L sections. TLO 1.5 Understand Moment of inertia for built-up section.	Unit - I Moment of Inertia 1.1 Concept of Moment of Inertia, M.I. of plane lamina and radius of gyration of a given lamina. 1.2 Parallel and perpendicular axes theorems (without derivation). 1.3 M.I. of standard basic figures like square, rectangle, triangle, circle, semi-circle, quarter-circle and Hollow Rectangular & Circular sections. (without derivation). 1.4 M.I. of Composite plane figures such as symmetrical and unsymmetrical I-section, channel section, T-section, angle section. Numerical on composite figure consisting of maximum 03 standard shapes. 1.5 Introduction to M.I. for built-up sections. (No numerical). (IKS* Concept of Centre of Gravity & M.I. used in ancient constructions like temples, forts etc.)	Chalk-Board Hands-on Collaborative learning Video Demonstrations Presentations

STRENGTH OF MATERIALS**Course Code : 313308**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Understand concept of stresses and strains in deformable bodies. TLO 2.2 Understand Hook's law, elastic limit, Linear and lateral strain, Poisson's ratio. TLO 2.3 Determine modulus of elasticity, modulus of rigidity and bulk modulus for given material. TLO 2.4 Articulate practical significance of stress- strain curve for given material under given loading conditions for their relevant use. TLO 2.5 Concept of single shear, double shear & punching shear. TLO 2.6 Compute the total deformation for given homogeneous (compound) sections under axial load. TLO 2.7 Determine the stresses in each material for given composite section. TLO 2.8 Compute strain along x, y and z-direction for a given bi-axial or tri-axial stress system.	Unit - II Simple Stresses, Strains & Elastic Constants 2.1 Concept of elastic body ,definition of stress, strain, Type of stresses & strains. 2.2 Hook's law, elastic limit, Linear and lateral strain, Poisson's ratio. 2.3 Young's Modulus, Shear Modulus, Bulk Modulus & Relation between these three moduli. 2.4 Standard stress strain curve for mild steel bar and Tor steel bar under tension test, Yield stress, proof stress, ultimate stress, breaking stress, and working stress, strain at various critical points, percentage elongation and Factor of safety. 2.5 Shear stress and shear strain, Single shear, Double shear, Punching shear. 2.6 Deformation of body subjected to axial force for uniformed and stepped sections .Deformation of uniform body subjected to forces at its intermediate sections. 2.7 Concept of composite section, stresses induced and load shared by each material under axial loading only.(No numerical on stepped sections). 2.8 Uni-axial, Bi-axial and Tri-axial stress systems. 2.9 Strain in each direction, volumetric strain, change in volume.	- Chalk-Board - Hands-on Collaborative learning - Video - Demonstrations - Presentations

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

STRENGTH OF MATERIALS**Course Code : 313308**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
	TLO 2.9 Determine volumetric strain & change in volume for given cube or cuboid.		
3	TLO 3.1 Enlist Types of Supports & Types of Beams TLO 3.2 Enlist types of loads acting on a beam. TLO 3.3 Understand the relation between SF, BM and rate of loading. TLO 3.4 Draw SFD and BMD for Simply supported beams, Cantilever beams & overhanging beams. TLO 3.5 Locate point of maximum BM and point of contra-flexure.	Unit - III Shear Force & Bending Moment 3.1 Types of Supports: Simple, Hinge, Roller & Fixed and Beams: Cantilever, Simply supported, Roller, Hinge & overhanging beams. 3.2 Types of loads: Concentrated or Point load, Inclined point load & Uniformly Distributed load. 3.3 Meaning of SF and BM, Relation between them, Sign conventions. 3.4 SFD & BMD for Simply Supported, Cantilever and overhanging beams subjected to Vertical point load & UDL only. 3.5 Drawing SFD and BMD, Location of Point of Contra-Shear, maximum BM, Location of Point of Contra-flexure.	Chalk-Board Hands-on Collaborative learning Video Demonstrations Presentations

STRENGTH OF MATERIALS**Course Code : 313308**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Understand concept of pure bending , Neutral Axis and radius of gyration of a given lamina and section modulus. TLO 4.2 Determine Moment of Resistance (M.R.) & section modulus (Z) using Flexural Formula. TLO 4.3 Determine the Bending stresses at given location in simply supported & cantilever beams subjected to standard loading cases (Point load & UDL only). TLO 4.4 Compute & draw maximum and average shear stress for rectangular and circular section. TLO 4.5 Draw shear stress distribution diagram for given section across its depth. TLO 4.6 Determine shear stresses in hollow rectangular section.	Unit - IV Bending and Shear Stresses in beams 4.1 Theory of pure bending, assumptions in pure bending, Concept of Neutral Axis and section modulus. 4.2 Flexural Equation (without derivation) with meaning of each term used in equation, bending stresses and their nature, bending stress distribution diagram. 4.3 Bending stress variation diagram across depth of given cross section for cantilever and simply supported beams for symmetrical sections only. 4.4 Shear stress equation (without derivation), meaning of each term used in equation, relation between maximum and average shear stress for square, rectangular and circular section (numerical), shear stress distribution diagram. 4.5 Shear stress distribution diagram for square, rectangular, circle, hollow square, hollow rectangular, hollow circle, T- section & symmetrical I- section only. (no numericals) 4.6 Use of shear stress equation for determination of shear stresses in hollow rectangular section.	Chalk-Board Hands-on Collaborative Learning Demonstration Video Presentations

STRENGTH OF MATERIALS**Course Code : 313308**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Explain effect of direct and eccentric loads on columns. TLO 5.2 Draw resultant stress distribution diagram for a compression member subjected to eccentric load about one of its principal axis. TLO 5.3 Write No tension condition for columns, Core of the section for rectangular & circular column. TLO 5.4 Identify the terms radius of gyration, slenderness ratio & effective length for given column with different end conditions. TLO 5.5 Understand the concept of buckling load in columns using Euler's Formula & Rankine's Formula.	Unit - V Direct and Bending Stresses 5.1 Introduction to direct and eccentric loads, Eccentricity about one principal axis, nature of stresses. 5.2 Maximum and minimum stresses, resultant stress distribution diagram. Condition for 'No tension' condition (Problems on Column subjected to Eccentric load about one axis only.) 5.3 Limit of eccentricity, core of section for circular cross sections, middle third rule for rectangular section. 5.4 Introduction to compression members, effective length, radius of gyration, slenderness ratio, type of end conditions for columns. 5.5 Buckling (or Crippling) load for columns by Euler's Formula & Rankine's Formula with meaning of each term in it. (No numericals.)	Chalk-Board Collaborative learning Presentations Demonstration Videos

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
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STRENGTH OF MATERIALS**Course Code : 313308**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify different components of UTM .	1	*Conduct sample compressive and tensile tests on metal using Universal Testing Machine along with introduction to machine & other tests to be conducted on UTM.	2	CO2
LLO 2.1 Perform Tension test on mild steel as per IS:432(1) .	2	*Tension test on mild steel as per IS:432(1) .	2	CO2
LLO 3.1 Perform tension test on Tor steel as per IS:1608, IS:1139 .	3	Tension test on Tor steel as per IS:1608, IS:1139 .	2	CO2
LLO 4.1 Conduct compression test on sample test piece using Compression Testing Machine .	4	*Compression test on any two metals like Mild Steel, Brass, Al etc. using Compression Testing Machine .	2	CO2
LLO 5.1 Conduct Izod Impact test on given metals as per IS:1598 .	5	*Izod Impact test on any two metals like mild steel/ brass/aluminum/ copper /cast iron etc. as per IS:1598 .	2	CO2
LLO 6.1 Conduct Charpy Impact test on given metals as per IS:1598 .	6	Charpy Impact test on two metals like mild steel/ brass/aluminum/ copper /cast iron etc. as per IS:1757 .	2	CO2
LLO 7.1 Determine Compressive strength of dry and wet bricks .	7	Compressive strength of dry and wet bricks as per IS:3495 (part I), IS:1077 .	2	CO2
LLO 8.1 Perform Single Shear and double shear test on given metals as per IS:5242 .	8	*Single Shear and double shear test on any two metals like Mild steel/ brass/ Al / copper / cast iron etc. as per IS:5242 .	2	CO2 CO4
LLO 9.1 Conduct Compression test on timber section along the grain and across the grain .	9	Compression test on timber section along the grain and across the grain as per IS:2408 .	2	CO1 CO2

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

STRENGTH OF MATERIALS**Course Code : 313308**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 10.1 Plot Shear force and Bending Moment diagrams of beams subjected to different types of loads.	10	*Shear force and Bending Moment diagrams of cantilever, simply supported and overhanging beams for different types of loading . (02 problems on each type of beam) .	4	CO3
LLO 11.1 Conduct Flexural test on timber beam on rectangular section.	11	*Flexural test on timber beam on rectangular section in both orientations as per IS:1708, IS:2408 .	2	CO1 CO4
LLO 12.1 Prepare PPT on Strain Energy. LLO 12.2 Prepare PPT on Thermal Stresses & Thermal Strains.	12	a) Prepare PPT of minimum 05 slides on the concept of Strain Energy & instantaneous stress induced in a material due to gradual, Sudden & impact load. b) Prepare PPT of minimum 04 slides on Thermal Stresses & Thermal Strains.	2	CO2
LLO 13.1 Conduct Flexure test on floor tiles/roofing tiles.	13	Flexure test on floor tiles IS:1237, IS:13630 or roofing tiles as per IS:654, IS:2690 .	2	CO4
LLO 14.1 Determine hardness no. for given metal using Rockwell Hardness Tester.	14	Rockwell Hardness Test on any two Metals like Mild Steel, Brass Copper, Aluminum etc.	2	CO2
LLO 15.1 Determine hardness no for given metals using Brinell Hardness Tester.	15	Brinell hardens test on any two metals like Mild Steel, Brass Copper, Aluminum etc.	2	CO2

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

STRENGTH OF MATERIALS**Course Code : 313308****VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)****Micro project**

- Collect the information of Indian Knowledge System (IKS) given in different units.
- Prepare charts of maximum bending moment and shear force values in standard beams.
- Two Numericals on Chimneys (of rectangular and circular cross section) subjected to wind pressure & also draw stress distribution diagram at base of it.
- Draw & identify difference between Bending stress distribution & Shear stress distribution diagrams for square, rectangular, circle, hollow square, rectangular, circle, T- section, & symmetrical I- section.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED**MSBTE Approval Dt. 02/07/2024****Semester - 3, K Scheme**

STRENGTH OF MATERIALS**Course Code : 313308**

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Extensometer with least count 0.01mm, maximum extension 25 mm with dial gauge/digital display suitable for various gauge length.	1,2,3
2	Accessories: Vernier caliper, meter scale, weighing balance, weights, punch, file, hammer, screw driver, pliers, etc.	1,2,3,4,5,6,7,8,9,11,13
3	Universal Testing Machine of capacity 1000kN, 600 kN / 400 kN, analog type/digital type with all attachments and accessories.	1,2,3,8,11,13
4	Tile flexural testing machine confirming to IS:654, capacity 200 Kg with uniform loading rate of 45 to 55 Kg/minute provided with lead shots .	13
5	Brinell and Rockwell Hardness Test machine .	14,15
6	Compression Testing Machine of capacity 2000 kN / 1000 kN, analog / digital type with all attachments and accessories.	4,7,9
7	Izod/Charpy impact testing machine confirming to IS: 1757.	5,6
8	Hot Air Oven with thermostatic control having temp. range 100 to 105 degree celsius .	7

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Moment of Inertia	CO1	10	2	4	6	12
2	II	Simple Stresses, Strains & Elastic Constants	CO2	16	6	8	4	18
3	III	Shear Force & Bending Moment	CO3	14	2	4	10	16
4	IV	Bending and Shear Stresses in beams	CO4	10	2	4	6	12
5	V	Direct and Bending Stresses	CO5	10	2	4	6	12

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

STRENGTH OF MATERIALS**Course Code : 313308**

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
Grand Total				60	14	24	32	70

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Two-unit tests of 30 marks each will be conducted and average of two-unit tests considered.
- For formative assessment of laboratory learning 25 marks
- Each practical will be assessed considering appropriate % weightage to process and product and other instructions of assessment.

Summative Assessment (Assessment of Learning)

- Pen and Paper Test (Written Test)

XI. SUGGESTED COS - POS MATRIX FORM

STRENGTH OF MATERIALS**Course Code : 313308**

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	3	2	2	1	--	2			
CO2	3	3	3	3	1	--	3			
CO3	3	3	2	1	1	--	2			
CO4	3	3	2	1	1	---	2			
CO5	3	3	2	1	1	--	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Khurmi R.S., Khurmi N.	A Textbook of Strength of Materials	S. Chand and Co. Ltd. New Delhi, 2019, ISBN 9789352833979
2	Ramamrutham S.	Strength of Materials	Dhanpat Rai and sons, New Delhi, 2015, ISBN 9788187433545

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

STRENGTH OF MATERIALS**Course Code : 313308**

Sr.No	Author	Title	Publisher with ISBN Number
3	Punmia B. C., Ashok Kumar Jain , Arun Kumar Jain .	Mechanics of Materials	Laxmi Publications (p) Ltd. New Delhi, 2017, ISBN-13: 978-8131806463
4	Rattan S.S.	Strength of Materials	McGraw Hill Education; New Delhi 2017, ISBN- 13: 978-9385965517
5	Rajput R. K .	A Textbook of Strength of Materials	S. Chand Publishing 9789352533695, 9352533690

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://youtu.be/DzyIEz3dKXQ?si=beGDRqJ1olZ70LUe	Concept of Stress and Strain .
2	https://youtu.be/RSImDKHDMUY? si=FHCxXE1QSaa0FqBn	Standard stress-strain curve for mild steel bar and Tor steel bar .
3	https://www.youtube.com/watch?v=MFZ18Ed4HI8	Field Test on TMT .
4	https://www.youtube.com/watch?v=C-FEVzI8oe8	Concept of SFD and BMD .
5	https://www.youtube.com/watch?v=yvbA4mk36Kk	Practical examples of SFD and BMD.
6	https://www.youtube.com/watch?v=f2eGwNUopws	Concept & Numerical on Point of Contraflexure .
7	https://www.youtube.com/watch?v=f08Y39UiC-o	Bending Stresses & Shear Stresses in Beams .
8	https://skyciv.com/structural-software/beam-analysis- softwar e	Calculation & Drawing of SFD & BMD freeware Software .

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

STRENGTH OF MATERIALS

Course Code : 313308

MSBTE Approval Dt. 02/07/2024

Semester - 3, K Scheme

GEOTECHNICAL ENGINEERING**Course Code : 314315**

Programme Name/s	: Civil Engineering/ Civil & Rural Engineering/ Construction Technology/ Civil & Environmental Engineering/
Programme Code	: CE/ CR/ CS/ LE
Semester	: Fourth
Course Title	: GEOTECHNICAL ENGINEERING
Course Code	: 314315

I. RATIONALE

The stability of any structure depends upon behavior of soil and bearing capacity of soil to carry loads under different loading conditions. Thus, the geotechnical engineering enables the decision maker to predict the behavior of soil under different loading conditions and also to determine the probable settlement arising from the construction activities. This course therefore will develop the basic understanding among the students to ensure the safety, stability, and long-term quality in the wide range of civil engineering projects such as buildings, dams, towers, embankments, roads, railways, retaining walls, bridges, underground tank and underwater structures.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Evaluate various soil properties required for design of foundation.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Apply the basic knowledge of Geology and Geotechnical Engineering in given situation
- CO2 - Measure the physical properties of given soil sample
- CO3 - Determine the shear strength of given soil sample
- CO4 - Use the relevant method of compaction to determine parameters of given soil sample

GEOTECHNICAL ENGINEERING**Course Code : 314315**

- CO5 - Undertake the relevant soil investigation techniques to determine the bearing capacity of the given soil strata

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL	FA-TH	SA-TH			Total		FA-PR		SA-PR		SLA				
Max	Max	Max	Min	Max	Min	Max	Min	Max	Min												
314315	GEOTECHNICAL ENGINEERING	GTE	DSC	3	-	2	3	8	4	3	30	70	100	40	25	10	25#	10	25	10	175

Total IKS Hrs for Sem. : 1 Hrs

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

GEOTECHNICAL ENGINEERING**Course Code : 314315****V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Elaborate the importance of geology in civil engineering field. TLO 1.2 Classify the given type of rocks based on their genesis. TLO 1.3 Signify the importance of soil as a construction material. TLO 1.4 Justify the importance of Geo-technical Engineering in civil Engineering.	Unit - I Overview of geology and geotechnical engineering 1.1 Introduction to Geology: Branches, importance of geology, composition of earth. 1.2 Petrology: Definition of a rock, classification based on their genesis (mode of origin), formation, classification and engineering uses of igneous, sedimentary and metamorphic rocks. (IKS*: Sun temple of Konark made up of Chlorite, Laterite, Khondalite stones) 1.3 IS definition of soil, Importance of soil in Civil Engineering as construction material for foundation bed of structures. 1.4 Field applications of geotechnical engineering for foundation design, pavement design, design of earth retaining structures, design of earthen dam.	Lecture Using Chalk-Board Presentations Demonstration Site/Industry Visit Case Study

GEOTECHNICAL ENGINEERING**Course Code : 314315**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Elaborate the physical properties of soil. TLO 2.2 Determine the Index properties of given soil sample using the relevant method. TLO 2.3 Draw the particle size distribution curve for the given sample with its interpretation. TLO 2.4 Interpret the computed values of Atterberg's limits of Consistency for the given soil specimen data.	Unit - II Physical and Index Properties of Soil 2.1 Physical Properties: Soil as a three phase system, water content, void ratio, porosity and degree of saturation, density index, unit weight of soil mass; bulk unit weight, dry unit weight, unit weight of solids, saturated unit weight, submerged unit weight, specific gravity 2.2 Determination of Index Properties of Soil: determination of water content by oven drying method as per IS code, determination of bulk unit weight and dry unit weight by core cutter method and sand replacement method as per IS code, determination of specific gravity by pycnometer. 2.3 Particle size distribution, mechanical sieve analysis as per IS code, particle size distribution curve, effective diameter of soil, Uniformity coefficient and coefficient of curvature, well graded and uniformly graded soils, particle size. classification of soils, I.S. classification of soil. 2.4 Consistency of soil: Stages of consistency, Atterberg's limits of consistency viz. Liquid limit, plastic limit and shrinkage limit, plasticity index, determination of liquid limit, plastic limit	Lecture Using Chalk-Board Presentations Demonstration Hands-on Site/Industry Visit

GEOTECHNICAL ENGINEERING**Course Code : 314315**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Apply the Darcy's law of permeability in given situation to determine the coefficient of permeability for a given soil sample. TLO 3.2 Apply the concept of flow net in the given situation. TLO 3.3 Draw the Mohr-coulomb failure envelope for the given type of soil sample. TLO 3.4 Use the relevant method to determine shear strength of given soil sample.	Unit - III Permeability and Shear Strength of Soil 3.1 Definition of permeability, Darcy's law of permeability, coefficient of permeability, factors affecting permeability, determination of coefficient of permeability by constant head and falling head permeability tests, simple problems to determine coefficient of permeability. 3.2 Seepage through earthen structures, seepage velocity, seepage pressure, phreatic line, flow lines, application of flow net, (No numerical problems.) 3.3 Shear failure of soil, field situation of shear failure, concept of shear strength of soil, components of shearing resistance of soil – cohesion, internal friction. Mohr-coulomb failure theory, Strength envelope, strength Equation for purely cohesive and cohesion less soils. 3.4 Laboratory methods: Direct shear test, vane shear test (Numerical on direct shear test only)	Lecture Using Chalk-Board Presentations Demonstration Hands-on Site/Industry Visit

GEOTECHNICAL ENGINEERING**Course Code : 314315**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Undertake the compaction of given sample using relevant method of compaction. TLO 4.2 Use the relevant method of soil stabilization for the given situation as per IS code. TLO 4.3 Determine the CBR value of given soil sample as per IS code. TLO 4.4 Explain the lateral earth pressure theory with labelled sketch for given situation.	Unit - IV Compaction and Stabilization of soil 4.1 Concept of compaction, purpose of compaction, field situations where compaction is required, Standard proctor test – test procedure as per IS code, Compaction curve, optimum moisture content, maximum dry density, Zero air voids line, Modified proctor test, factors affecting compaction, field methods of compaction : rolling, ramming and vibration, concept of consolidation, difference between compaction and consolidation. 4.2 Concept of soil stabilization, necessity of soil stabilization 4.3 California bearing ratio, C.B.R. test, interpretation of C.B.R. values. 4.4 Definition of earth pressure, lateral earth pressure at rest, active earth pressure and passive earth pressure with no surcharge condition, coefficient of earth pressure, Rankine's theory and its assumptions.	Lecture Using Chalk-Board Presentations Video Demonstrations Hands-on Site/Industry Visit

GEOTECHNICAL ENGINEERING**Course Code : 314315**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Undertake relevant Exploration Technique for evaluating soil strata. TLO 5.2 Determine the bearing capacity of soil using the relevant data for the given soil sample. TLO 5.3 Justify the need of field test in determining the bearing capacity of the soil for the given strata	Unit - V Site Investigation and Bearing Capacity of Soil 5.1 Site Investigation: Necessity of site investigation and sub-soil exploration, types of exploration, criteria for deciding the location and number of test pits and bores. Field identification of soil: dry strength test, dilatancy test and toughness test, Determination of free swell index. 5.2 Bearing capacity: Definition of bearing capacity, ultimate bearing capacity, safe bearing capacity and allowable bearing pressure, Introduction to Terzaghi's analysis and its assumptions (No Numerical). Types of failures in soil: general, local and punching shear failure, effect of water table on bearing capacity. 5.3 Field methods for determination of bearing capacity – Plate load test and standard penetration test. Test procedures as Per IS: 1888 & IS:2131	Lecture Using Chalk-Board Presentations Video Demonstrations Collaborative learning Site/Industry Visit

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify the type of given rock specimen.	1	*Identification of rocks from the given specimen	2	CO1
LLO 2.1 Use oven drying method to determine the percentage of moisture content in given soil specimen.	2	*Determination of moisture content of given soil sample by oven drying method as per I.S. 2720 part-II	2	CO2

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

GEOTECHNICAL ENGINEERING**Course Code : 314315**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 3.1 Use pycnometer method for determining specific gravity of given soil sample to classify its type.	3	*Determination of specific gravity of soil by pycnometer method as per I.S. 2720 part- III.	2	CO2
LLO 4.1 Undertake the core cutter method to find bulk and dry unit weight of given soil sample in field.	4	*Determination of Bulk and dry unit weight of soil in field by core cutter method as per I.S. 2720 (Part- XXIX).	2	CO2
LLO 5.1 Undertake the sand replacement method to find bulk and dry unit weight of coarse-grained soils.	5	Determination of bulk and dry unit weight of soil in field by sand replacement method as per I.S. 2720 (Part- XXVIII).	2	CO2
LLO 6.1 Classify type of soil based on grain size distribution.	6	*Determination of grain size distribution of given soil sample by mechanical sieve analysis as per I.S. 2720 (Part- IV).	2	CO2
LLO 7.1 Identify given soil based on plasticity chart.	7	*Determination of Plastic Limit & Liquid Limit along with Plasticity Index of given soil sample as per I.S. 2720 (Part- V).	2	CO2
LLO 8.1 Find co efficient of permeability of soil through coarse grained soils and fine grained soil.	8	*Determination of co efficient of permeability by constant head test as per I.S. 2720 (Part- XVII) or Determination of co efficient of permeability by falling head test as per I.S. I.S. 2720 (Part- XVII)	2	CO3
LLO 9.1 Determine the shear strength of soil sample using direct shear test.	9	Determination of shear strength of soil by direct shear test as per I.S. 2720 (Part-XIII)	2	CO3
LLO 10.1 Determine shear strength of soil in undisturbed as well as remoulded cohesive soil sample.	10	Determination of shear strength of soil by vane shear test as per I.S. 2720 (Part-XXX)	2	CO3

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

GEOTECHNICAL ENGINEERING**Course Code : 314315**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 11.1 Perform standard proctor test to determine parameters such as OMC, MDD and amount of compaction	11	*Determination of OMC and MDD by standard proctor test of given soil sample as per I.S. 2720 (Part- VII).	2	CO4
LLO 12.1 Perform modified proctor test to determine parameters such as OMC, MDD and amount of compaction	12	Determination of OMC and MDD by Modified proctor test of given soil sample as per I.S. 2720 (Part- VIII).	2	CO4
LLO 13.1 Perform the CBR test on a given soil sample to evaluate thickness of pavement	13	Determination of CBR value as per IS 2720 (Part- 16).	2	CO4
LLO 14.1 Classify given soil sample by conducting field tests Through Visual inspection, Dry strength test, Dilatancy test and Toughness test.	14	Use of field tests to identify type of given soil sample.	2	CO5
LLO 15.1 Find degree of Expansiveness of given soil based on free swell index.	15	Determination of free Swell index of soil as per IS 2720 (Part 40)	2	CO5
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)/ASSIGNMENTS

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

GEOTECHNICAL ENGINEERING**Course Code : 314315**

- a. Collect the data of various rock specimen such as igneous, sedimentary, metamorphic and compare their properties with respect to geotechnical Engineering.
- b. Collect minimum three pictures rock mass showing Folds, Faults, Joints along with description.
- c. Observe open source videos for determination of Shrinkage limit of soil sample as per I.S. 2720 (Part- V) and write a procedure.
- d. Determine bearing capacity of soil using assumed value of liquid limit and plastic limit and show calculations (Workout bearing capacity of soil using established co relation.)
- e. Enlist minimum five compaction equipments along with their description
- f. Enlist various soil stabilization techniques and write information of any three (including sketches).
- g. Comment in the form of report on the effect of unconsolidated undrained, consolidated undrained and consolidated drained in shear test
- h. Summarize the importance of Geosynthetic materials including their applications in civil Engineering
- i. Summarize in the form report on the importance of piles including sketches and case studies
- j. Write a report on role of Geophysical Exploration in civil Engineering.

Micro project

- a. Collect minimum five types of rock specimen in your area and compare their properties with respect to geotechnical Engineering aspect
- b. Visit nearby site having excavation pits and write short note about it strata by visual inspection.
- c. Collect minimum three samples of soil in your area and compare them for any three properties of soil
- d. Visit nearby reclamation land and study the stabilization method.
- e. Identity two different locations and suggest the appropriate stabilization methods (soil-cement, soil-lime, soil-flyash etc.) to improve its engineering properties.
- f. Study different free open sources software available for Geotechnical Engineering.
- g. Collect the photograph and information of anchors (stabilization of slopes) used to avoid over turning of structure.
- h. Collect the photographs and information on Causes, Effects and Types of Landslides.

GEOTECHNICAL ENGINEERING**Course Code : 314315**

- i. Collect information on foundations of ancient structures with Geotechnical Engineering aspect.
- j. Visit nearby two sites to classify soil based on field tests.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Vane shear test apparatus- as per 2720 (Part -30)	10
2	Proctor compactometer for light compaction and heavy compaction as per IS specification	11,12
3	CBR apparatus as per IS specification IS 2720 (Part-16).	13
4	425 micron IS Sieve and 100ml capacity graduated glass cylinder	15
5	Oven-thermostatically controlled to maintain temperature of 110 degree Celsius to 115 degree Celsius	2,3,4,7,11,12

GEOTECHNICAL ENGINEERING**Course Code : 314315**

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
6	Pycnometer – consisting of 1 kg. honey /fruit jar with plastic cone, locking ring and rubber seal.	3
7	Core cutter apparatus- cylindrical core cutter of steel 100 mm dia x 127.3mm high with 3mm wall thickness beveled at 1mm.	4
8	Sand replacement apparatus- as per IS: 2720(Part-28)	5
9	Mechanical sieve shaker- carries up to 7 sieves of 15 cm to 20 cm dia (as per IS 2720-(Part 4)1985)	6
10	Casagrande liquid limit apparatus- as per IS: 9259-1979	7
11	Constant head permeameter- as per IS:2720(Part-4)1986	8
12	Falling head permeameter -as per IS:2720(Part-4)1986	8
13	Direct shear test apparatus- as per IS: 2720(Part 13) 1986	9

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Overview of geology and geotechnical engineering	CO1	5	4	4	0	8
2	II	Physical and Index Properties of Soil	CO2	12	4	4	12	20
3	III	Permeability and Shear Strength of Soil	CO3	10	2	8	6	16
4	IV	Compaction and Stabilization of soil	CO4	10	4	4	6	14
5	V	Site Investigation and Bearing Capacity of Soil	CO5	8	0	8	4	12
Grand Total				45	14	28	28	70

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

GEOTECHNICAL ENGINEERING**Course Code : 314315****X. ASSESSMENT METHODOLOGIES/TOOLS****Formative assessment (Assessment for Learning)**

- Two unit test of 30 marks will be conducted and average of two unit test is considered, Assessment of laboratory learning , Assignment, Microproject, Self learning (60% Weightage to process and 40% weightage to product), Question and Answer.

Summative Assessment (Assessment of Learning)

- Pen and Paper Test (Written Test), Practical Exam/ Oral Exam

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	-	-	2	1	-	2			
CO2	3	3	1	3	2	1	3			

GEOTECHNICAL ENGINEERING**Course Code : 314315**

CO3	2	2	1	2	1	2	2			
CO4	2	2	2	2	2	1	2			
CO5	2	2	2	2	2	1	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Punmia, B.C.	Soil Mechanics and Foundation Engineering	Laxmi Publication (P) ltd., New Delhi, ISBN 9788170087915
2	Murthy, V.N.S.	A text book of soil mechanics and foundation Engineering	CBS Publishers & Distributors Pvt. Ltd., New Delhi 2016 ISBN: 9788123913629
3	Ramamurthy, T.N. & Sitharam, T.G.	Geotechnical Engineering (Soil Mechanics)	S Chand and Company LTD., New Delhi, ISBN: 9788121924573
4	Braja M. Das	Principles of Geotechnical Engineering	Cengage Learning ISBN: 9789355738103
5	Parbin Singh	Engineering And General Geology	S K Kataria and Sons ISBN-13 978-8188458516

XIII. LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://youtu.be/N2J-tvEeI4c?si=SgQP0ICSbFAuOVLd	Determination of Water Content of Soil by Oven Drying Method
2	https://youtu.be/l6vk0EM4yPg?si=-M6WwrxOlZYHy94	Determination of Specific Gravity of soil

GEOTECHNICAL ENGINEERING**Course Code : 314315**

Sr.No	Link / Portal	Description
3	https://youtu.be/5rDHjZ_RJq0?si=V714qwz1vE8f5pSP	Determination of Dry Density of Soil by Core Cutter Method
4	https://youtu.be/YejCi5SEOAU?si=n8w1uAauI7ZgaG9P	Determination of Dry Density of Soil by Sand Replacement Method
5	https://youtu.be/pM-w_cvk1nA?si=3evWNLPjtwFxtsP0	Determination of Liquid Limit and Plastic Limit of Soil
6	https://youtu.be/bmpn5oNDvOs?si=LxcoQUSe_lmL3QQ1	Direct Shear Test
7	https://youtu.be/CAezS3mPzOc?si=9-bIMPqTKy1MuPDG	Grain Size Analysis of Soil
8	https://youtu.be/c4i_y6u-tsE?si=BjcItf55LqNn2Ihn	Water Content Dry Density Relation Using Light Compaction OMC and MDD
9	https://youtu.be/fCmMW73rP64?si=mdAiq1WPkpc9n1Dl	California Bearing Ratio (CBR) value test
10	https://youtu.be/Lrml0egYtM4?si=ag1mezmk74UAuyCf	Determination of swelling properties (Free Swell Index of Soil)
11	https://smfe-iiith.vlabs.ac.in/List%20of%20experiments.html	Virtual laboratory practical for Soil Mechanics.
12	https://youtu.be/8Q8CZW9-jXE?si=8Yrf2NvS9b5v9kcF	A Soil Investigation Work (Borehole Drilling: SPT & Rock Coring)
13	https://www.ijsrp.org/research-paper-0121/ijsrp-p10935.pdf	The Architectural Study of Sun Temples in India: Based on Location, Construction Material and Spatial Analysis Study
14	https://youtu.be/QuE4tEK-5iY?si=t61uZOWhS_nd5z8H	Shallow Foundation: Plate Load Test

GEOTECHNICAL ENGINEERING**Course Code : 314315**

Sr.No	Link / Portal	Description
15	https://youtu.be/DjWDOqQjsyQ?si=k43rXl2I19YK9msV	How to conduct SPT / Standard Penetration Test/ Soil Exploration Technique/ Site Investigation
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

MSBTE Approval Dt. 21/11/2024**Semester - 4, K Scheme**

BUILDING PLANNING & DRAWING WITH CAD

Course Code : 313009

Programme Name/s	: Civil Engineering/ Civil & Rural Engineering/ Construction Technology/ Civil & Environmental Engineering/
Programme Code	: CE/ CR/ CS/ LE
Semester	: Third
Course Title	: BUILDING PLANNING & DRAWING WITH CAD
Course Code	: 313009

I. RATIONALE

Building planning and drawing are critical components of the construction process, enabling stakeholders to visualize, communicate, problem-solve, comply with regulations, estimate costs, guide construction, and ensure quality throughout the project lifecycle. In today's era of globalization and technology revolutions, it is necessary to prepare the civil engineering drawings in such a way that it can be prepared with utmost precision and accuracy with ability to modify it as and when required. This is only possible if the said drawing is prepared using the CAD software. This course is therefore planned with the goal of developing such competency among the learners.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Prepare Drawings of the given building structure with required specifications using CAD Software.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Draw the line plans of given type of buildings considering the Principles of Planning.
- CO2 - Use CAD software for drafting and editing of the given type of drawing.
- CO3 - Draw the relevant type of plan/drawings for the given type of building.
- CO4 - Draw perspective drawing for the given type of objects.

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

- CO5 - Draw the Isometric and 3- Dimensional drawings of the given component of the structure.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
															Max	Min	Max	Min	Max	Min	
313009	BUILDING PLANNING & DRAWING WITH CAD	BDC	SEC	2	-	4	2	8	4	-	-	-	-	-	50	20	50#	20	25	10	125

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009****Total IKS Hrs for Sem. : 1 Hrs**

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
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BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain the given basic principles of Planning of building (residential and public). TLO 1.2 Propose the relevant dimensions for the given component of building structures. TLO 1.3 Plan the dwelling unit as per the given requirement and specifications TLO 1.4 Plan the dwelling units/building in accordance with the provisions of governing authority in a given area TLO 1.5 Compute the required area of construction using the norms of the local authority. TLO 1.6 Draw line plans for the given residential buildings. TLO 1.7 Draw line plans for the given public buildings.	Unit - I Principles of Planning 1.1 Principles of planning of Residential and Public building: Aspect, Prospect, Orientation, Grouping, Privacy, Elegance, Flexibility, Roominess, Circulation, Furniture requirements, Sanitation, Economy. (IKS-Orientation of Indian Heritage Structures such as Mahalaxmi Temple, Kolhapur) 1.2 Space requirement and norms for minimum dimension of different components of building structure. 1.3 Planning of residential buildings as per the given requirement using IS 962-1989. 1.4 Rules and bye-laws of sanctioning authorities (local authority and town planning department) for construction work. 1.5 Plot area, built up area, super built up area, plinth area, carpet area, floor area and FAR (Floor Area Ratio) / FSI. 1.6 Line plans for residential building of minimum three rooms including w/c, bath and staircase as per principles of planning 1.7 Line plans for public building-primary health center, restaurant, bank, post office, hostel and Library.	Video Demonstrations Presentations Site/Industry Visit Case Study Lecture Using Chalk-Board

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Explain the various software available with its importance for drawing through CAD to be used in the given situation. TLO 2.2 Undertake the required initial settings of the relevant software to draw the new drawing. TLO 2.3 Use the relevant command to draw the specific feature of the figure or plan in given situation. TLO 2.4 Modify the given drawing/figure using relevant command in given situation. TLO 2.5 Calculate the parameters like distance, area, perimeter of a given figure. TLO 2.6 Trace the anchor point on the given boundaries of the figure to get the relevant dimension of the figure. TLO 2.7 Use the concept of layer to visualize the specific component of the building/drawing. TLO 2.8 Apply the required dimension command to mark the	Unit - II Computer aided drawing 2.1 Fundamentals : various drafting software for civil engineering applications. System requirement for drawing software. Advantages of computer aided drawing over traditional method of drawing. 2.2 Initial setting required to start new drawing. 2.3 Draw commands: Line, poly line, construction line, rectangle, polygon, circle, ellipse, hatch, boundary, text, arc, point, make block. 2.4 Modify commands: Erase, copy, mirror, offset, trim, move, extend, rotate, array, lengthen, scale, chamfer, fillet, explode, stretch. 2.5 Compute area of the given drawing 2.6 Changing properties of entity: line type, color, scale, font- size, color, style. 2.7 Layer command: Create layers for given components of given drawing. 2.8 Dimension toolbar: Quick dimension, linear dimension, and continuous dimension; align dimension, angular dimensions, dimension style. 2.9 Use of plot/print command for the output of given drawing.	Video Demonstrations Presentations Hands-on Model Demonstration Lecture Using Chalk-Board

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
	dimensions in the given drawing. TLO 2.9 Use the output of the in the required format using specific command.		
3	TLO 3.1 To prepare Submission Drawing of load bearing and Framed structure in accordance with the provisions of governing authority in a given area. TLO 3.2 Prepare working drawing of the given Load bearing and Framed structure TLO 3.3 Prepare foundation plan of the given Load bearing and Framed structure. TLO 3.4 Prepare structural drawings of given RCC Components of the building structure.	Unit - III Planning of Buildings 3.1 Data drawing (for Load bearing and framed structure): Developed plan, elevation, section, site plan, schedule of openings, construction notes with specifications, area statement. Planning of staircase- Rise and Tread for residential building. 3.2 Working drawing: Developed plan, elevation, section passing through staircase or Sanitary Block. 3.3 Foundation plan of Load bearing and Framed structure. 3.4 Details of RCC Components: Footing, column, Beam, Chajjas, Lintel, Staircase and slab.	Video Demonstrations Demonstration Model Demonstration Site/Industry Visit Lecture Using Chalk-Board

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Explain the principles of perspective drawings in the given situation TLO 4.2 Prepare perspective drawing of the given object using Two-point perspective method	Unit - IV Perspective Drawing 4.1 Definition, Types of perspective, terms and principles used in perspective drawing 4.2 Two Point Perspective of objects- steps, monuments, pedestals.	Video Demonstrations Hands-on Lecture Using Chalk-Board Model Demonstration
5	TLO 5.1 Justify the need of isometric drawing in civil engineering TLO 5.2 Prepare 3: dimensional drawing using the relevant 3 Dimensional Modify Commands.	Unit - V Introduction to 3-Dimensional drawings 5.1 Isometric drawing: meaning and necessity, use of isometric snap, isometric axes, isocircle, isotext 5.2 3-Dimensional drawing: Use of, Extrude, Press pull, Union, Subtraction command for preparing drawing of components-Beam, Columns etc.	Video Demonstrations Hands-on Lecture Using Chalk-Board Demonstration

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Represent the given door/window section using their standard signs and symbols.	1	* Draw various types of graphical symbols for materials, doors and windows as per IS 962:1989 (Sketch book)	2	CO1

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 2.1 Use the symbols to different water supply and Sanitary units of the building in the drawing	2	Draw various types of graphical symbols for sanitary, water supply as per IS 962:1989 (Sketch book) .	2	CO1
LLO 3.1 Use the symbols to different Electrical units of the building in the drawing.	3	Draw various types of graphical symbols for electrical installations and write abbreviations as per IS 962:1989 (Sketch book)	2	CO1
LLO 4.1 Illustrate different specifications of the given drawings.	4	Write summary of observations of all technical details from the given drawing (One/Two BHK) obtained from the professional architect or civil engineer (Sketch book)	2	CO1
LLO 5.1 Identify the different units of the building and prepare the plan as per the actual measurements.	5	*Measure the units of existing building (Load Bearing / Frame structure) & Draw line plan of measured existing building-(Part - I) (Sketch book) .	2	CO1
LLO 6.1 Identify the different units of the building and prepare the plan as per the actual measurements.	6	* Measure the units of existing building (Load Bearing / Frame structure) & Draw line plan of measured existing building-(Part - II) (Sketch book) .	2	CO1
LLO 7.1 Prepare Line Plans of the Residential building as per the requirement.	7	* Draw line plan to suitable scale (Minimum 1BHK, staircase, WC and Bathroom) for Residential Bungalows.(Minimum three) (Sketch book)	2	CO1
LLO 8.1 Prepare Line Plans of the Public building as per the requirement.	8	Draw line plans to suitable scale for Public Buildings -Primary Health Centre (Sketch book)	2	CO1

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 9.1 Prepare Line Plans of the Public building as per the requirement.	9	Draw line plans to suitable scale for Public Buildings – Hostel, Library. (Sketch book)	2	CO1
LLO 10.1 Prepare Line Plans of the Public building as per the requirement.	10	*Draw line plans on sketch book to suitable scale for Public Buildings- Bank, Post Office - (Sketch book)	2	CO1
LLO 11.1 Prepare Developed Plan of the residential building as per the requirement.	11	Draw the Developed plan and Elevation for a Framed Structure (One/Two BHK)-Part I (Sketch book)	2	CO3
LLO 12.1 Prepare Section of the residential building as per the requirement.	12	Draw the Section through WC, Bath or Staircase for Plan in Practical No.11 for a Framed Structure (One/Two BHK) -Part II (Sketch book)	2	CO3
LLO 13.1 Prepare Site Plan and area statement of the residential building as per the requirement.	13	Draw the Site plan including area statement, schedule of opening and construction notes for a Framed Structure (One/Two BHK) - for Plan in Practical No.11 Part III (Sketch book)	2	CO3
LLO 14.1 Prepare Perspective view of the given object.	14	*Draw two-point perspective drawing of small objects - steps, monuments, pedestals (anyone) with suitable scale (Sketch book)	2	CO4
LLO 15.1 Prepare Line Plans of the Residential building as per the requirement.	15	Prepare Line plan of Residential Building with staircase (minimum two) using CAD Software. (A-2 Size Half Imperial Sheet)	2	CO2

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 16.1 Prepare Line Plans of the Public building as per the requirement.	16	Prepare Line plan of Public Building with staircase (minimum Two) with using CAD Software. (A-2 Size Half Imperial Sheet)	2	CO2
LLO 17.1 Prepare Plan and sectional elevation of the Staircase as per the requirement.	17	*Draw plan and Sectional Elevation of dog-legged staircase using CAD Software. (A-2 Size Half Imperial Sheet)	2	CO2
LLO 18.1 Prepare Developed Plan of the load bearing structure as per the requirement.	18	*Draw submission drawing to suitable scale of a single storey load bearing residential building (2BHK) with flat Roof and staircase showing Developed plan and elevation using CAD Software.-Part I (A-2 Size Half Imperial Sheet)	2	CO3
LLO 19.1 Prepare Sectional elevation of the load bearing structure as per the requirement.	19	*Draw submission drawing to Suitable scale of a single Storey load bearing residential building (2BHK) with flat Roof and staircase showing - Section passing through Stair or W.C. and Bath using CAD Software - for Plan in Practical No.18 -Part II (A-2 Size Half Imperial Sheet)	2	CO3
LLO 20.1 Prepare Foundation Plan of the load bearing structure as per the requirement.	20	*Draw submission drawing to the suitable scale of a single Storey load bearing residential building (2BHK) with flat Roof and staircase showing - a) Foundation plan and schedule of openings. b) Site plan with suitable scale, area statement, construction notes for Plan in Practical No.18- using CAD Software -Part III- using CAD Software. (A-2 Size Half Imperial Sheet)	2	CO3

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 21.1 Prepare Developed Plan of the Framed structure as per the requirement.	21	*Draw submission drawing, to the suitable scale of (G+1) Framed Structure Residential Building (2BHK) with Flat Roof and staircase showing: a) Developed plan b) Elevation - Part I Using CAD Software (A-2 Size Half Imperial Sheet)	2	CO3
LLO 22.1 Prepare Sectional elevation of the framed structure as per the requirement.	22	*Draw submission drawing, to the suitable scale of (G+1) Framed Structure Residential Building (2BHK) with Flat Roof and staircase showing. - Section passing through Stair, W.C. and Bath- For plan in Practical No.21 - Part II-using CAD Software (A-2 Size Half Imperial Sheet)	2	CO3
LLO 23.1 Prepare site plan and area statement of the framed structure as per the requirement.	23	*Draw submission drawing, to the suitable scale, of (G+1) Framed Structure Residential Building (2BHK) with Flat Roof and staircase showing - a) Site plan and area statement b) Schedule of openings and construction notes - For plan in Practical No.21 - Part III using CAD Software (A-2 Size Half Imperial Sheet)	2	CO3
LLO 24.1 Prepare working drawing the framed structure as per the requirement.	24	Draw Foundation with suitable scale for Practical No.21 using CAD Software (A-2 Size Half Imperial Sheet)	2	CO2
LLO 25.1 Prepare Sectional elevation and plan of footing for framed structure as per the requirement.	25	Draw Detailed enlarge section of RCC column and footing with suitable scale using CAD Software. (A-2 Size Half Imperial Sheet)	2	CO2

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 26.1 Prepare Sectional elevation and plan of RCC Beam, Lintel Beam and Chajja of framed structure as per the requirement.	26	*Draw Detailed enlarge section of RCC Beam, Lintel and Chajja with suitable scale using CAD Software. (A-2 Size Half Imperial Sheet)	2	CO2
LLO 27.1 Prepare Sectional elevation and plan of Stair case as per the requirement.	27	Draw Detailed enlarge section of RCC staircase with suitable scale using CAD Software. (A-2 Size Half Imperial Sheet)	2	CO2
LLO 28.1 Prepare Isometric drawings of given object.	28	*Draw isometric drawing of simple objects –Straight Lines and edges (minimum 02 objects) using CAD Software. (A-2 Size Half Imperial Sheet)	2	CO5
LLO 29.1 Prepare Isometric drawings of given object.	29	Draw isometric drawing of simple objects-Curved Edges (minimum 02 objects) using CAD Software. (A-2 Size Half Imperial Sheet)	2	CO5
LLO 30.1 Prepare 3-D view of given object.	30	*Draw 3-D view of simple object. (any one) - using CAD Software. (A-2 Size Half Imperial Sheet)	2	CO5
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

BUILDING PLANNING & DRAWING WITH CAD

Course Code : 313009

Micro project

- Collect and study building Bye laws, rules and regulation for planning any two competent authority such as Gram-Panchayat/Municipal Corporation/Metro Cities/Town Planning Department.
- Prepare report on Provisions given in National Building Code 2005
- Prepare list of the documents required for obtaining permission for construction of residential building/apartment from competent authority and write report.
- Prepare a report on IS-962:1989 - Code of practice for architectural and building drawings
- Prepare Developed Plan and Elevation for a any one Public Building using CAD Software.

Self Learning

- List any five software's used for building planning and drawing and prepare one developed plan using any one Free opensource software.
- Prepare a model of a simple building using cardboard showing different components with suitable color.
- Prepare line plan as per given requirement Using CAD Software. (Any THREE -- other than mentioned in curriculum)
- Free Online Courses on Auto CAD by NPTEL /Coursera/IGNOU/SWAYAM

Assignment

- State and explain the classification of residential buildings with respect to Planning such as Row house/Apartments/detached /Semi-detached Buildings
 - Explain the Role of Architect, Structural Engineer and Supervisor in Planning of Building.
 - Prepare a report on BUILDING PLAN MANAGEMENT SYSTEM -By Urban Development Department Government of Maharashtra.
 - Prepare a report on Building Plan Approval Process as per NBC -2005
-

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009****Note :**

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Computer with specification as 8GB RAM, Graphics Card 4 GB, HDD/SSD 500GB, LCD Monitor with relevant CAD software. (with latest configuration)	All
2	Printer preferably for the output of A-3 size, paper	All
3	LCD projector.	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Principles of Planning	CO1	6	0	0	0	0

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
2	II	Computer aided drawing	CO2	4	0	0	0	0
3	III	Planning of Buildings	CO3	12	0	0	0	0
4	IV	Perspective Drawing	CO4	4	0	0	0	0
5	V	Introduction to 3-Dimensional drawings	CO5	4	0	0	0	0
Grand Total				30	0	0	0	0

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Term work, Assignment, Microproject (60% Weightage to process and 40% weightage to product)

Summative Assessment (Assessment of Learning)

- Practical Exam, Oral Exam

XI. SUGGESTED COS - POS MATRIX FORM

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	02	01	02	01	02		02			
CO2	02	02	02	03	02	01	02			
CO3	02	02	02	02	02		02			
CO4	02	01	02	02	02		02			
CO5	02	01	01	03	02		01			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Shah. M.G., Kale C.M., Patki S.Y.	Building Drawing	Mcgraw Hill Publishing company Ltd. New Delhi 2002 ISBN: 9780074638767
2	Dr. Swamy Kumara N; Rao Kameshwara A .	Building Planning and Drawing	Charotar Publication, ANAND ISBN : 978-93-85039-12-6 (Ed.2015)

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

BUILDING PLANNING & DRAWING WITH CAD**Course Code : 313009**

Sr.No	Author	Title	Publisher with ISBN Number
3	Mantri Sandip	A to Z Building Construction	Satya Prakashan; 2nd edition (2015), New Delhi, ISBN: 978-8176849692
4	Nighat Yasmin Ph.D.	Introduction to AutoCAD 2024 for Civil Engineering Applications	SDC Publications, ISBN: 978-1-63057-607-3
5	Malik R.S., Meo G.S.	Civil Engineering Drawing	Computech Publication Ltd New Asian Publishers, 2009, New Delhi ISBN:978-8173180026

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.youtube.com/watch?app=desktop&v=E6TE9u1XgAg	2D Commands - NITTTR Chandigadh-NCTEL
2	https://www.youtube.com/watch?v=rX6XfCMRYU0	Demonstration video 2-Point Perspective View-Basics.
3	https://www.youtube.com/watch?app=desktop&v=N4FUbpGAWNA	3D Commands in Autocad - NITTTR Chandigadh -- NCTEL
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**